

# Force Sensing and Compliance Control for a Cable-Driven Redundant Manipulator

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**Abstract**—The cable-driven redundant manipulator (CDRM) has a light slender and highly dexterous body, which is especially suitable for fine manipulation in confined spaces or unstructured environments. However, for dexterous manipulation and physical interaction with its surrounding environment, the force/torque sensor is indispensable, which will significantly increase the mass, dimension, and cost of the whole CDRM system. In this article, a force-sensing algorithm and compliance control framework for CDRM without a six-axis force/torque sensor are proposed. First, we design a modular cable-driven manipulator with two-level internal sensors, i.e., joint encoders and cable tension sensors. The multispace kinetic model is derived to establish the mapping between motor, cable, joint, and end-effector states. At the same time, we build a recursive dynamics model that takes the cables' tension, cable-hole friction, links' gravity, and end-effector forces into account. Then, the indirect force-sensing algorithm of the end-effector is proposed by combining the kinematic and dynamic equations and internal sensor information. Furthermore, a compliance controller based on indirect force sensing is designed. Finally, typical experiments are carried out based on the CDRM. Experiment results indicate that the force-sensing accuracy exceeds 95%, whereas the compliance controller demonstrates outstanding compliant behavior in human-robot interaction tasks.

**Index Terms**—Admittance control, cable-driven redundant manipulator (CDRM), dynamic model, force sensing, human-robot interaction.

## I. INTRODUCTION

THE cable-driven manipulator has great advantages, such as a slender and continuous body, self-compliance, flexible motion, which are inspired by animals [1]. With these particular features, cable-driven manipulators are able to operate in narrow spaces [2], [3]. A wide range of practical applications have been presented in the literature, including minimally invasive surgery [4], [5], toxic chemical and nuclear waste manipulation [6], [7], search and rescue [8], and inspection in space or other domains [9]. Moreover, a redundant manipulator has more directly or indirectly driven degrees of freedom (DOF) [10] than industrial robots [11].

In the last several decades, the dynamic models of single and multisections of cable-driven manipulators have been explored using various methods, such as the Euler-Lagrange formalism [12], the Cosserat rod theory [13], [14], the Newton-Euler formalism [15], and the principle of virtual work (PVW) [16]. In these studies, some nonlinear equations need to be solved using the nonlinear programming solver, which limits the dynamic model's applicability in real time [17]. Ammar et al. [18] proposed a dynamic model that disregards the effects of friction, which may significantly minimize the time required for numerical calculations but compromises the model's correctness. Consequently, the friction could not be ignored, and many studies comparing friction models found that the Coulomb model is less accurate than the others [19], [20].

In order to sense and interact with the environment, the cable-driven manipulator needs to have force-sensing capability [21]. The force-sensing methods can be divided into direct and indirect methods. Direct methods usually rely on different sensors, such as fiber Bragg grating sensors [21], [22], which are placed between the cover and the link of the manipulator [23] or the six-axis force/torque ( $F/T$ ) sensor. On the other hand, the studies on indirect force sensing mostly used *ex situ* force sensing and took the estimation of end-effector force by the PVW as intrinsic force sensing. In addition, Xu et al. [24] used screw theory to analyze the limitations and provided a geometric interpretation for the intrinsic force sensing. Furthermore, the team of Simaan used joint-level *ex situ* force sensing in addition [25]. Previous methods for force sensing in tube robots used

Manuscript received 21 November 2022; revised 6 February 2023; accepted 25 March 2023. Date of publication 19 April 2023; date of current version 16 February 2024. Recommended by Technical Editor S. Y. Yoon and Senior Editor W. J. C. Zhang. This work was supported in part by the National Natural Science Foundation of China under Grants 62233001 and 62203140, in part by the Program of Shenzhen Peacock Innovation Team under Grant KQTD20210811090146075, in part by the Shenzhen Excellent Scientific and Technological Innovation Talent Training Project under Grant RCJC20200714114436040 and Grant RCBS20221008093122054, and in part by the Basic Research Program of Shenzhen under Grant JCYJ20220818102415034. (Corresponding authors: Lei Yan; Bin Liang.)

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This article has supplementary material provided by the authors and color versions of one or more figures available at <https://doi.org/10.1109/TMECH.2023.3263922>.

Digital Object Identifier 10.1109/TMECH.2023.3263922

indirect force-sensing methods with errors ranging from 5% to 20% of maximum reported loads (tip forces at the level of 1 N) [26]. These methods greatly enhance the ability of force sensing, but two factors that are still being ignored are gravity and friction, especially at larger loads. In addition, a similar cable-driven manipulator could not achieve compliance control if the force sensing has large errors or noise.

For cable-driven manipulators, the hysteretic nonlinear system is usually compensated by friction as feedforward to design open-loop controllers such as Qi et al. [27]. Other researchers utilized neural networks to avoid the complexity of modeling nonlinear dynamics [28]. However, the strategy adopted for the closed-loop controller, whether model-based or model-free, depends on the feedback of force or joint angle sensors [19], [32]. Although cable-driven manipulators with joint encoders have been presented in several research studies [30], [31], [32], the compliance control and the human–robot interaction (HRI) have received less attention. This is likely owing to the difficulties posed by the superior force-sensing technique and compliance controller design. In addition, the research studies on the compliance control of the cable-driven manipulators and continuum robots have stalled at the level of numerical simulation and are unable to effectively implement the HRI or teaching–playback applications.

In this article, we focus on two challenges related to the viability of force sensing and compliance control in robotic systems with nonlinear and robust coupling.

**Challenge 1: Limits of the model-based dynamic model.** In general, the volume limitation of the manipulator makes it particularly difficult to arrange all the force sensors and encoder sensors. On the other hand, because of the existence of angle errors, tension sensor errors, and cable–hole friction, it is difficult to accurately sense the external force.

**Challenge 2: Indirectly driving to achieve compliance control.** Compared with an integrating robot joint, the cable-driven redundant manipulator (CDRM) could not control torque directly in the joints, so it could not realize impedance control. In addition, the cost and weight of six-axis  $F/T$  are also limitations for the controller design.

In this article, considering the above-mentioned challenges, we deliver the modified dynamic model and propose a novel hybrid cable–hole friction model. Furthermore, we have overcome challenges such as system design and integration, ex situ force sensing, and compliance controller design, which also realize the HRI. Hence, the main contributions of this article are summarized as follows.

- 1) We propose an indirect force-sensing method, which is realized by combining kinematics, modified dynamics, and internal sensor information, including joint encoders and cable tension sensors. The indirect force-sensing method can reduce the cost of CDRM and significantly improve its reliability and safety.
- 2) We first design a compliance controller for CDRM based on the force-sensing method, which can be used for physical HRI and teaching–playback applications. The related experimental results show the outstanding performance of the proposed controller.

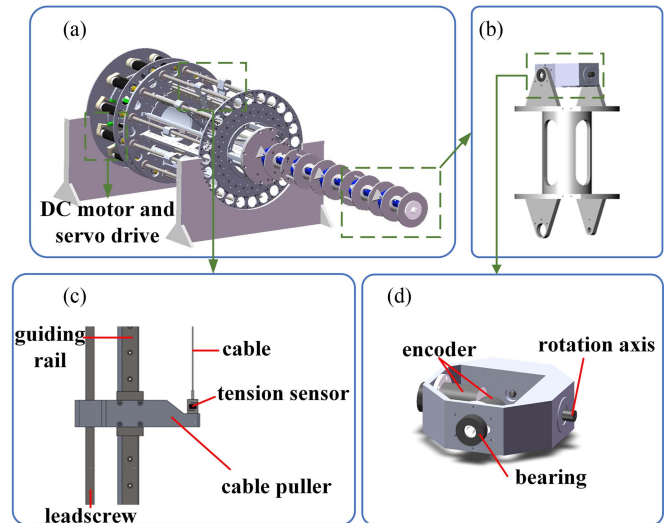


Fig. 1. Prototype of the CDRM. (a) Robotic system overview. (b) One link section. (c) One driving unit. (d) One joint section.

The rest of this article is organized as follows. In Section II, the structure and kinematics of CDRM are introduced, including robot system design and kinematics modeling. In Section III, we propose the force-sensing and dynamic model considering friction. In Section IV, we design the admittance control law and controller in detail. We carry out four typical experiments in Section V. Finally, Section VI concludes this article.

## II. SYSTEM DESIGN AND KINEMATIC MODELING

### A. System Design

The CDRM system needs to guarantee high motion accuracy and safety, which means it needs to have obstacle avoidance and heavy payload abilities. According to the above-mentioned requirements, we have completed the design of the mechanical structure, including the designs of the universal joint, modular link, and drive box, as shown in Fig. 1.

The prominent advantage of the abovementioned structural design is electromechanical separation. The primary role of a drive box is to protect the structures and other hardware for easy maintenance and disassembly. Each cable needs a motor unit to make it stretch, and therefore, each dc motor drives the linear movement of the cable through the coupling, leadscrew, guiding rail, tension sensor, and cable puller mechanism, as shown in Fig. 1(c).

The modular link structure is the main body of the moving part of the CDRM system, and the universal joint has been set up between two links, which has 2 DOFs of pitch–yaw, as shown in Fig. 1(b) and (d). Considering that signal lines and other electrical lines need to be placed in the center of the link, the prototype will be designed as a hollow structure to reserve space.

The universal joint is the core component of the orthogonal installation of the two links and also undertakes the function of measuring the angle values of the joints, as shown in Fig. 1(d). The radial load of the rotary shaft of the absolute magnetic

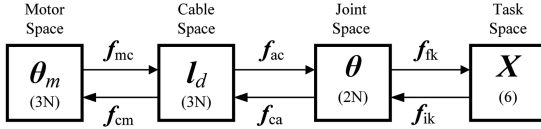


Fig. 2. Multispace mapping relationship of motor-cable-joint-task.

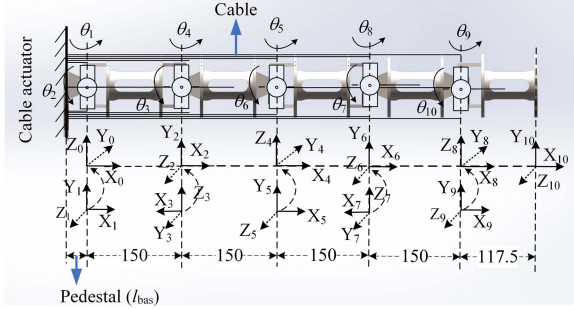


Fig. 3. D-H framework.

encoder is limited and cannot be directly used as the rotary shaft. Therefore, a coaxial structure is designed such that the link can rotate while the encoder and the center stay relatively still, so as to realize the measurement of the joint angle.

### B. Simply Kinematic Modeling

The CDRM is composed of a series of links and synchronously drives the motion of each joint through multiple cables. Therefore, besides the mapping relationship between the end-effector and the joint angle of the manipulator, a mapping relationship between the joint angle value and the cable length of the driving space should also be established. Therefore, the kinetic mapping of the manipulator has four parts, as shown in Fig. 2.

In forward kinematics (FK), the Denavit-Hartenberg (D-H) method is used to recursively solve the fixed coordinate system of each rotation pair according to the real-time joint angle of the manipulator, until the end-effector pose is solved [1].

For the 10-DOF redundant manipulator designed in this article, many scholars have studied a variety of inverse kinematics (IK) solutions to establish the mapping relationship from the end-effector pose to the joint angles.

The IK based on the Jacobian pseudoinverse numerical iterative solution is adopted in this article, and this method was first proposed by Whitney [33]. The iterative process is repeated until the error between the end-effector pose of the manipulator and the expected pose is smaller than the predefined threshold value, at which point the calculation is finished. The closer the initial pose is to the expectation, the less the number of iterations is.

## III. FORCE SENSING BASED ON THE DYNAMIC MODEL

### A. Dynamic Modeling

Since CDRM control is a nonlinear and strongly coupled system, we must first build an effective dynamic model in order to improve control performance. The Newton-Euler dynamics

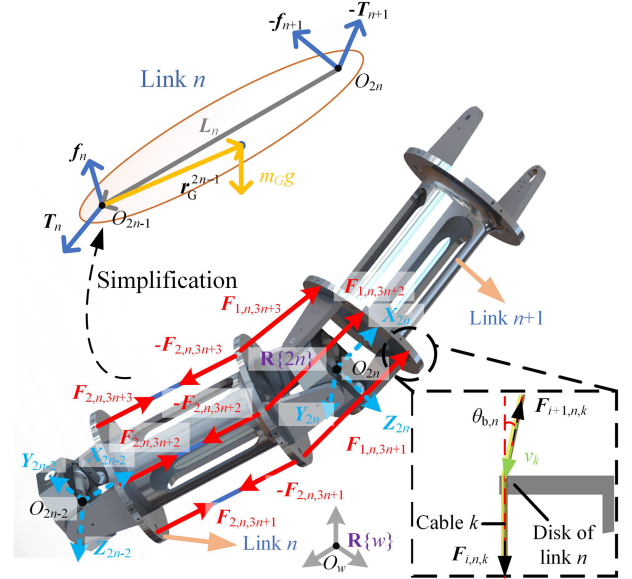


Fig. 4. Cable force, moment, and friction analysis of double links.

model is a method that describes the dynamics of a single rigid body and then solves the whole robotic system. In general, the obtained result is a set of equations composed of multiple equations. Based on our previous research [1], we propose a novel hybrid friction model to modify the dynamic model, which would decrease the error of the force-sensing method at the same time.

First, the coordinate frame is constructed according to the classical D-H method to determine the local coordinate frame  $\mathbf{R}\{X_{2n}, Y_{2n}, Z_{2n}\}$  of two universal joints, as shown in Fig. 4.  $f_{n+1}$  and  $T_{n+1}$  represent the external force and torque exerted by link  $n$  at link  $n+1$ , respectively, and are similar to  $f_n$  and  $T_n$ .  $r_G$  is the position vector of the centroid of the connecting rod relative to the coordinate system  $2n$ , and  $L_n$  is the position vector of joint  $n$  relative to the coordinate system  $2n$ . Furthermore, it is worthy of note that due to the intersecting cross-axis structure of the universal joint, the coordinate frame origin of the joint coincided with the coordinate frame of the last link.

The position vector of the cable-hole in the coordinate frame  $\mathbf{R}\{2n\}$  should be obtained before the tension analysis of the cable of a single link. According to the analysis presented in Fig. 4, the position vector  $p_{n,k}^{2n}$  of the cable-hole relative to the coordinate frame  $\mathbf{R}\{2n\}$  can be obtained according to the following formula:

$$p_{1,n,k}^{2n} = \begin{cases} [d-l & rs\beta_k & -rc\beta_k] & n=1,3\dots \\ [d-l & rc\beta_k & rs\beta_k] & n=2,4\dots \end{cases}$$

$$p_{2,n,k}^{2n} = \begin{cases} [-d & rs\beta_k & -rc\beta_k] & n=1,3\dots \\ [-d & rc\beta_k & rs\beta_k] & n=2,4\dots \end{cases} \quad (1)$$

where  $d$  is the distance between two surfaces of a single universal joint,  $l$  is the link's length,  $\beta_k$  is the angle between the cable-hole of  $k$  and the  $y$ -axis based on  $\mathbf{R}\{2n\}$ , and  $s$  and  $c$  represent  $\sin()$  and  $\cos()$ , respectively. Meanwhile, a single cable on one link



has three vectors, which are given as follows:

$$\begin{cases} \mathbf{L}_{1,n,k}^0 = \mathbf{R}_{2n+2}^0 \mathbf{p}_{2,n+1,k}^{2n+2} - \mathbf{R}_{2n}^0 \mathbf{p}_{1,n,k}^{2n} \\ \mathbf{L}_{2,n,k}^0 = \mathbf{R}_{2n}^0 \mathbf{p}_{1,n,k}^{2n} - \mathbf{R}_{2n}^0 \mathbf{p}_{2,n,k}^{2n} \\ \mathbf{L}_{3,n,k}^0 = \mathbf{R}_{2n}^0 \mathbf{p}_{2,n,k}^{2n} - \mathbf{R}_{2n-2}^0 \mathbf{p}_{1,n-1,k}^{2n-2} \end{cases} \quad (2)$$

where the  $\mathbf{R}_{2n-1}^0$  is the rotation matrix of  $\mathbf{H}_{2n-1}^0$ . Therefore, the force vector is obtained by calculating

$$\mathbf{F}_{i,n,k}^0 = \mathbf{F}_{i,n,k} \mathbf{L}_{i,n,k}^0 / \|\mathbf{L}_{i,n,k}^0\| \quad i = 1, 2, 3. \quad (3)$$

However, the measurement of the tensile force sensor is not the acting force  $F_{i,n,k}$  due to the influence of frictional force. For this reason, we propose a hybrid friction model to build the cable-hole dynamic friction model rather than single Coulomb friction in this article [19], [20]. In the hybrid friction model, there are generally four kinds of friction: Coulomb friction, static friction, viscous friction, and Stribeck friction, and the expression formula of cable  $k$  friction  $f_{i,k}$  is given as follows:

$$f_{i,k} = \begin{cases} F_k(v) & v \neq 0 \\ F_e & v = 0 \text{ and } F_e < F_{S,k} \\ F_{S,k} \operatorname{sgn}(F_e) & \text{other} \end{cases} \quad (4)$$

where  $F_{S,k}$  is the maximum static friction force between solids, and  $F_e$  is relative force. On one hand, the static friction force is related to the value of external force at a relatively stationary configuration. On the other hand, since the relationship between friction and velocity is not linear and there is a phenomenon of static being converted to dynamic in the process of moving, the value of dynamic friction is related to many parameters. Therefore, the general nonlinear calculation formula of dynamic friction is expressed as follows:

$$F_k(v) = F_{C,k} + (F_{S,k} - F_{C,k})e^{-|v_k/v_\sigma|^{\delta_\sigma}} + F_{V,k}v_k \quad (5)$$

where  $F_{C,k}$  is the Coulomb friction,  $F_{V,k}$  is the proportional coefficient of viscous friction,  $v_k$  is the relative speed of the cable, and  $k$ ,  $v_\sigma$ , and  $\delta_\sigma$  are Stribeck velocity and Stribeck coefficient, respectively. Due to the contact between the cable and the hole, the cable will bend at the end of the hole when passing through, as shown in Fig. 4.

The other part of (5) is composed of Coulomb friction, which reflects the proportional relationship between friction and load force and is obtained as follows:

$$F_{C,k} = \mu f_{N,k} \operatorname{sgn}(v_k) \quad (6)$$

where  $\mu$  is the coefficient of saturated viscous friction,  $f_{N,k}$  is the applied force,  $v$  is the relative velocity, and  $\operatorname{sgn}()$  is the sign function to return the positive or negative sign. The coefficient of friction  $\mu$  changes with different angles, and the applied force  $f_{N,k}$  is less than the tension force  $\mathbf{F}_{i,n,k}$  (bending angle  $< 30^\circ$ ). Hence, the Coulomb force  $F_C$  can be estimated as follows:

$$\begin{aligned} F_{C,k} &= \mu(\theta_{b,n}) \|\mathbf{F}_{i,n,k} + \mathbf{F}_{i+1,n,k}\| \operatorname{sgn}(v_k) \\ &= \mu'(\theta_{b,n}) F_{i,n,k} \operatorname{sgn}(v_k) \quad i = 1, 2, \dots \end{aligned} \quad (7)$$

where  $\mu'$  is the equivalent coefficient as a function of the bending angle  $\theta_{b,n}$  of the universal joint  $n$ . According to Liu et al. [34], the relationship between the  $\theta_{b,n}$  and universal joint angle  $\theta_{2n-1}$ ,

$\theta_{2n}$  is given as follows:

$$\begin{cases} \theta_{b,n} = \begin{cases} \operatorname{atan2} \left( \begin{matrix} -\sin \theta_{2n} \sin \beta_n, \\ \cos \theta_{2n-1} \cos \theta_{2n} \sin^2 \beta_n \end{matrix} \right) & (\sin \beta_k \neq 0) \\ \operatorname{atan2} \left( \begin{matrix} \sin \theta_{2n-1} \cos \theta_{2n} \cos \beta_n, \\ \cos \theta_{2n-1} \cos \theta_{2n} \cos^2 \beta_n \end{matrix} \right) & (\sin \beta_k = 0) \end{cases} \\ \beta_n = \operatorname{atan2}(-\sin \theta_{2n}, \sin \theta_{2n-1} \cos \theta_{2n}) \end{cases} \quad (8)$$

The other part of (5) represents viscous friction, which results from the viscous phenomenon caused by fluid layers between solid contact surfaces. Viscous friction is usually affected by factors such as velocity and bending angle. Considering that the bending angle is not constant and is always less than  $30^\circ$  and the viscous friction changes with  $\theta_{b,n}$  linearly, we express  $F_{V,k}$  as follows:

$$F_{V,k} = F_v(\theta_{b,n})v_k = K_v \theta_{b,n}v_k \quad (9)$$

where  $K_v$  is the viscous coefficient. The Stribeck friction  $F_{sbk}$  effect reflects the phenomenon that the friction force decreases as the speed increases at low speeds, and the relative velocity of motion between solids is related to the maximum static friction force. In order to simplify the calculation, we regard the maximum static friction force  $F_{S,k}$  as a constant  $S_{fm}$  and ignore the influence of cable velocity as follows:

$$F_{sbk} = (S_{fm} - F_{C,k})e^{-|v_k/v_\sigma|^{\delta_\sigma}}. \quad (10)$$

In general, the formula for calculating the dynamic friction force of the cable-hole based on the classical friction theory is given as follows:

$$\begin{aligned} F_{i+1,n,k} &= \mu'(\theta_{b,n}) F_{i,n,k} \operatorname{sgn}(v_k) + K_v \theta_{b,n}v_k \\ &+ (S_{fm} - \mu'(\theta_{b,n}) F_{i,n,k} \operatorname{sgn}(v_k)) e^{-|v_k/v_\sigma|^{\delta_\sigma}}. \end{aligned} \quad (11)$$

When the cable is loose, the tension gradually increases from the drive box to the end-effector, and the friction force can be calculated according to (11). In another situation, when the cable is taut, the tension can be obtained as follows:

$$F_{i+1,n,k} = \frac{F_{i,n,k} - K_v \theta_{b,n}v_k - S_{fm} e^{-|v_k/v_\sigma|^{\delta_\sigma}}}{\mu'(\theta_{b,n}) (1 - e^{-|v_k/v_\sigma|^{\delta_\sigma}})}. \quad (12)$$

Besides the friction and cable forces, the cable generated active and indirect moments for the single link section, respectively. According to (1)–(12), we can obtain the active moment  $M_{act}^{2n-1}$  in frame  $\mathbf{R}\{X_{2n-1}, Y_{2n-1}, Z_{2n-1}\}$  as follows:

$$\mathbf{M}_{act}^{2n-1} = \mathbf{R}_{2n-1}^0 \sum_{k=3n-2}^{3n} (\mathbf{p}_{2,n,k}^0 \times \mathbf{F}_{3,n,k}^0) \quad (13)$$

where  $\mathbf{p}_{2n,k}$  is the hole position vector. In the same way, indirect moment  $M_{ind}^{2n-1}$  and total cable moment  $M_{cable}^{2n-1}$  are derived as follows:

$$\mathbf{M}_{ind}^{2n-1} = \mathbf{R}_{2n-1}^0 \sum_{k=3n+1}^{3N} \left( \mathbf{p}_{2,n,k}^0 \times (\mathbf{F}_{2,n,k}^0 - \mathbf{F}_{3,n,k}^0) + \mathbf{p}_{1,n,k}^0 \times (\mathbf{F}_{1,n,k}^0 - \mathbf{F}_{2,n,k}^0) \right) \quad (14)$$

$$\mathbf{M}_{cable}^{2n-1} = \mathbf{M}_{act}^{2n-1} + \mathbf{M}_{ind}^{2n-1}. \quad (15)$$



Similarly, gravitational moment  $M_G^{2n-1}$  depends on the gravity position vector  $r_G^{2n-1}$  and gravity  $G_L^{2n-1}$  of link

$$M_G^{2n-1} = r_G^{2n-1} \times G_L^{2n-1}. \quad (16)$$

According to the force balance equation, there is also force transfer  $f_{n+1}$  (see Fig. 4) between adjacent links. Hence, the transmission moment  $M_{\text{link}}^{2n-1}$ , which is generated by the connecting link, can be written as follows:

$$\begin{aligned} M_{\text{link}}^{2n-1} &= L_n^{2n-1} \times f_{n+1}^{2n-1} + T_{n+1}^{2n-1} \\ &= L_n^{2n-1} \times \left( \sum_{i=1}^3 F_{i,n,k}^0 + G_L^{2n-1} + F_{\text{ext}}^{2n-1} \right) + T_{n+1}^{2n-1}. \end{aligned} \quad (17)$$

There are 2 DOFs at the universal joint, and thus the y-axis and z-axis moments are balanced with the cable moment. Therefore, the transmission moment  $T_{n+1}^{2n-1}$  of the front link must be parallel to the x-axis

$$T_{n+1}^{2n-1} = [\tau_{n+1} \ 0 \ 0]. \quad (18)$$

Through the aforementioned equations and moment balance equation, we can obtain the total external moment  $M_{\text{total}}$  as follows:

$$M_{\text{total}} = M_{\text{cable}} + M_G + M_{\text{link}} + M_{\text{Fext}} \quad (19)$$

where  $M_{\text{Fext}}$  is the external force acting at the end-effector. The inertial force of the universal joint effects accounts for CDRM's resistance to changing the linear and angular velocities. The absolute magnetic encoder can measure the angle information in real time; furthermore, the Newton–Euler method is adopted to calculate the angular velocity and angular acceleration of the link and the joint as follows:

$$\omega_{L_n}^{2n} = \begin{cases} R_0^2 z \dot{\theta}_1 + R_1^2 z \dot{\theta}_2 & (n=1) \\ R_{2n-2}^{2n} \omega_{L_{n-1}}^{2n-2} + R_{2n-2}^{2n} z \dot{\theta}_{2n-1} + R_{2n-1}^{2n} z \dot{\theta}_{2n} & (n>1) \end{cases} \quad (20)$$

$$\alpha_{L_n}^{2n} = \begin{cases} R_0^2 z \ddot{\theta}_1 + R_0^2 z \dot{\theta}_1 \times R_1^2 z \dot{\theta}_2 + R_1^2 z \ddot{\theta}_2 & (n=1) \\ R_{2n-2}^{2n} (\omega_{L_{n-1}}^{2n-2} \times z \dot{\theta}_{2n-1}) + R_{2n-2}^{2n} z \ddot{\theta}_{2n-1} \\ + R_{2n-2}^{2n} \dot{\omega}_{L_{n-1}}^{2n-2} + R_{2n-1}^{2n} z \ddot{\theta}_{2n} \\ + R_{2n-2}^{2n} ((z \dot{\theta}_{2n-1} + \omega_{L_{n-1}}^{2n-2}) \times R_{2n-1}^{2n} z \dot{\theta}_{2n}) & (n>1) \end{cases} \quad (21)$$

$$\omega_{B_n}^{2n-1} = \begin{cases} R_0^1 z \dot{\theta}_1 & (n=1) \\ R_{2n-2}^{2n-1} (z \dot{\theta}_{2n-1} + \omega_{B_{n-1}}^{2n-3}) & (n>1) \end{cases} \quad (22)$$

$$\alpha_{B_n}^{2n-1} = \begin{cases} R_0^1 \ddot{\theta}_1 & (n=1) \\ R_{2n-2}^{2n-1} (\alpha_{B_{n-1}}^{2n-3} + \omega_{B_{n-1}}^{2n-3} \times z \dot{\theta}_{2n-1} + z \ddot{\theta}_{2n-1}) & (n>1) \end{cases} \quad (23)$$

where  $\omega_{L_n}^{2n}$  and  $\alpha_{L_n}^{2n}$  are the angular velocity and angular acceleration of link  $n$ , respectively, in frame  $\mathbf{R}\{2n\}$ , and  $\omega_{B_n}^{2n}$  and  $\alpha_{B_n}^{2n}$  are the angular velocity and angular acceleration of universal

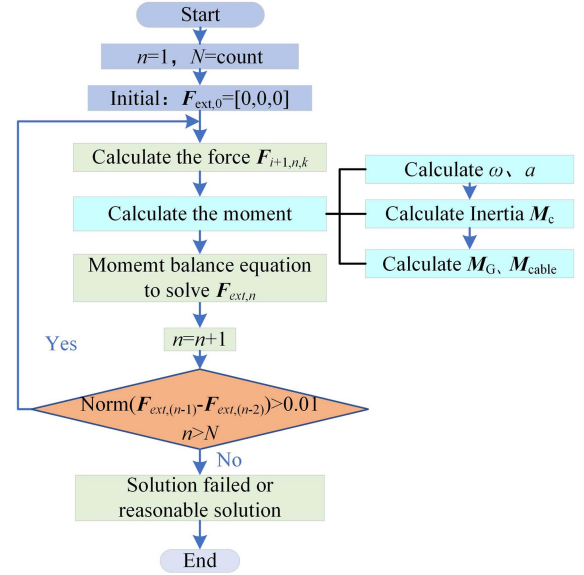


Fig. 5. Steps for the numerical solution of force sensing. We set the initial sensing value  $F_{\text{ext},0} = [0 \ 0 \ 0]$  and iteratively calculate the force until the error is within the 0.01 N.

joint  $n$ , respectively, in frame  $\mathbf{R}\{2n-1\}$ . Furthermore, the total moment of inertia  $M_c^{2n-1}$  is given as follows:

$$\begin{aligned} M_C^{2n-1} &= I_L^{2n-1} \alpha_L^{2n-1} + \omega_L^{2n-1} \times I_L^{2n-1} \omega_L^{2n-1} \\ &+ I_B^{2n-1} \alpha_B^{2n-1} + \omega_B^{2n-1} \times I_B^{2n-1} \omega_B^{2n-1} \end{aligned} \quad (24)$$

where  $I_L^{2n-1}$  and  $I_B^{2n-1}$  are the radial moments of inertia in frame  $\mathbf{R}\{2n-1\}$ . The magnitude of the interaction moment between the links is unknown, and the direction is parallel to the normal vector of the universal joint's surface. Therefore, the dynamic model balance equation only considers the y- and z-axes of frame  $\mathbf{R}\{2n-1\}$ , and the dynamic balance equation of a single joint is given as follows:

$$R_{\text{am}} M_C^{2n-1} = R_{\text{am}} M_{\text{total}}^{2n-1} \quad (25)$$

and  $R_{\text{am}}$  is the auxiliary matrix, which is given as follows:

$$R_{\text{am}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (26)$$

## B. Force-Sensing Method

The working environment of the CDRM is often complex and narrow, and end-effector six-axis  $F/T$  sensors will increase load and cost. Hence, we propose a novel force-sensing method for CDRM to sense the environment and avoid colliding with other obstacles.

This ex situ force-sensing method combines the dynamic model of CDRM with the dynamic friction force model of the cable-hole. It mainly includes the steps shown in Fig. 5.

The force analysis of the center block subsystem of the end link is carried out as shown in Fig. 6. The x-axis component is not calculated in this step because the end force cannot generate a moment in the frame  $\mathbf{R}\{2n\}$ .

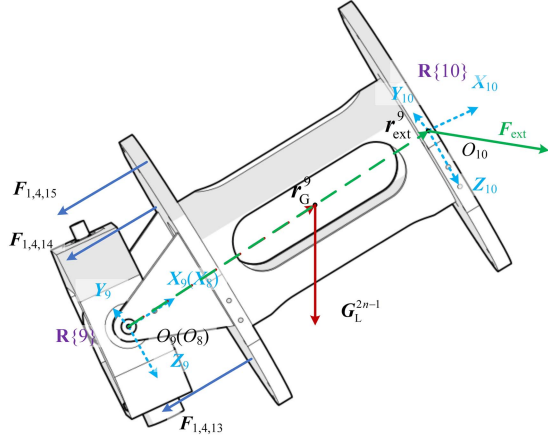


Fig. 6. Forces and moments analysis of end link.

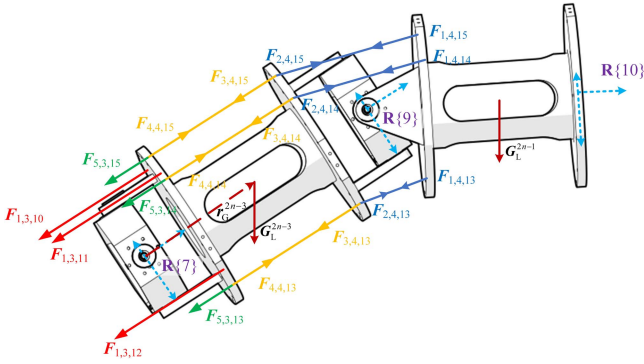


Fig. 7. Force analysis of end link and  $N-1$  link.

From the cable moment (15), moment balance (25), and moment of inertia (24), the following can be obtained:

$$\mathbf{R}_{\text{am}} (\mathbf{M}_{\text{cable}}^{2n-1} + \mathbf{M}_{\text{G}}^{2n-1} + \mathbf{r}_{\text{ext}}^{2n-1} \times \mathbf{F}_{\text{ext},1}) = \mathbf{R}_{\text{am}} \mathbf{M}_{\text{C}}^{2n-1} \quad (27)$$

where  $\mathbf{r}_{\text{ext}}^{2n-1} = [l \ 0 \ 0]$  is the position vector of external force.  $\mathbf{F}_{\text{ext},1}$  can be obtained from all the above-mentioned equations without the  $x$ -axis component and is given as follows:

$$\mathbf{F}_{\text{ext},1} = [0 \ F_{y0} \ F_{z0}]^T. \quad (28)$$

Obviously, a single iteration process reduces the precision and capability of force sensing. Therefore, we need to perform force analysis on the  $N-1$  link in order to improve accuracy.

From (13)–(15), we can derive the cable moment  $\mathbf{M}_{\text{cable}}^{2n-3}$ . Similar to the end joint, according to the total inertia moment  $\mathbf{M}_{\text{C}}^{2n-3}$ , we can obtain the moment balance equation as follows:

$$\begin{aligned} \mathbf{R}_{\text{am}} (\mathbf{M}_{\text{cable}}^{2n-3} + \mathbf{M}_{\text{G}}^{2n-3} + \mathbf{r}_{\text{ext}}^{2n-3} \times \mathbf{F}_{\text{ext},1} + \mathbf{r}_{\text{ext}}^{2n-3} \times \mathbf{F}_{x0}) \\ = \mathbf{R}_{\text{am}} \mathbf{M}_{\text{C}}^{2n-3} \end{aligned} \quad (29)$$

where  $\mathbf{F}_{x0}$  is the  $x$ -axis component of the sensing force  $\mathbf{F}_{\text{ext}}$  in the  $\mathbf{R}\{2n\}$ . Hence, the sensing force  $\mathbf{F}_{\text{ext},2}$  of the second iteration is expressed as follows:

$$\mathbf{F}_{\text{ext},2} = [F_{x0} \ F_{y0} \ F_{z0}]^T. \quad (30)$$

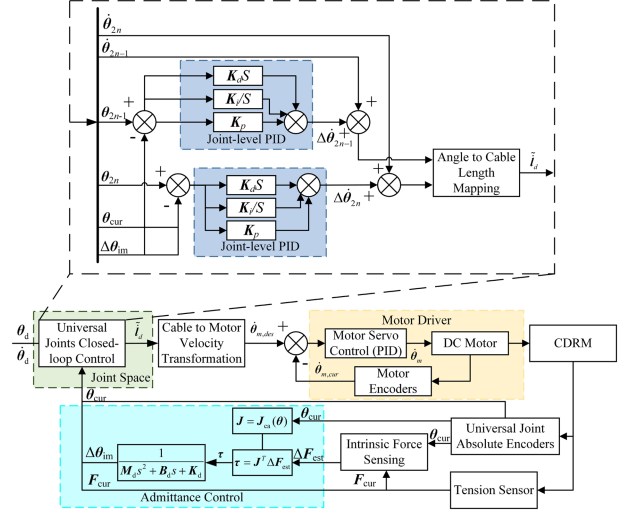


Fig. 8. Admittance control framework, including force sensing and the other low-level controllers.

The principle of iteration is the same for the other subsegments, and the significant differences in the calculation are the inertia moment  $\mathbf{M}_{\text{C}}$  and gravitational moment  $\mathbf{M}_{\text{G}}^{2n-3}$ . Repeat the above-mentioned steps until the error in sensing force value  $\mathbf{F}_{\text{ext}}$  is less than the limitation or the number of solutions exceeds the number of link  $N$ ; at that point, the iteration process is ended.

#### IV. COMPLIANCE CONTROL

##### A. Force Sensing and Admittance Control Frame

The admittance control calculates the change in the joint angle of the manipulator through the torque of each joint, as shown in Fig. 8.

The input of the admittance control is the external force information obtained by force sensing, and the output is the joint angles. The control law is given as follows:

$$\mathbf{M}(\ddot{\boldsymbol{\theta}}_{\text{d}} - \ddot{\boldsymbol{\theta}}_{\text{cur}}) + \mathbf{D}(\dot{\boldsymbol{\theta}}_{\text{d}} - \dot{\boldsymbol{\theta}}_{\text{cur}}) + \mathbf{K}(\boldsymbol{\theta}_{\text{d}} - \boldsymbol{\theta}_{\text{cur}}) = \boldsymbol{\tau} \quad (31)$$

where  $\mathbf{M}$ ,  $\mathbf{D}$ , and  $\mathbf{K}$  represent the mass, damping, and stiffness coefficients, respectively,  $\boldsymbol{\theta}_{\text{d}}$  is the desired joint angle, and  $\boldsymbol{\theta}_{\text{cur}}$  is the current joint angle. As shown in Fig. 8, after detecting the change in the external force at the end-effector of the CDRM through the force-sensing algorithm, the force value is converted into a universal equivalent joint torque  $\boldsymbol{\tau}$ , which can be obtained by the Jacobian matrix. The formula is solved by the bilinear transform method, and the transfer function of the linear time-invariant system in the continuous-time domain is converted to a discrete-time domain transfer function. The transfer function is obtained as follows:

$$\Delta\theta_{\text{im}}(n_T) = \frac{\tau(n) + 2\tau(n-1) + \tau(n-2) - a_1\theta_{\text{cur}}(n-1) - a_2\theta_{\text{cur}}(n-2)}{a_0} \quad (32)$$

where  $\Delta\theta_{\text{im}}(n_T)$  is the deviation of the angle for the  $n_T$  control cycle, and  $a_0$ ,  $a_1$ , and  $a_2$  are polynomials, which are given as

follows:

$$\begin{cases} a_0 = 4MT^{-2} + 2DT^{-1} + K \\ a_1 = -8MT^{-2} + 2K \\ a_2 = 4MT^{-2} - 2DT^{-1} + K \end{cases} \quad (33)$$

where  $T$  is the control cycle. The angle deviation value for each control cycle is calculated using the aforementioned equations, and it is then added to the joint angle command as an input of the closed-loop controller to achieve admittance control.

The universal joint closed-loop controller combines the advantages of feedforward and feedback control. There is also a mapping between joint velocity and cable velocity in CDRM. A single universal joint has two DOFs, and each DOF requires a controller that outputs joint velocity commands. The relationship between the motor speed  $\dot{\theta}_m$  and the cable speed  $\dot{i}_d$  is given as follows:

$$\dot{\theta}_m = f_{cm}(\dot{i}_d) = \frac{c_g \dot{i}_d}{S_{scr}} \quad (34)$$

where  $c_g$  is the reduction ratio of the gearbox and  $S_{scr}$  is the ball screw lead. As a result of coupling effects, the relative absolute speed of the cable is superimposed on the other joints and is given as follows:

$$\dot{i}_d = \sum_{i=1}^{2N} f_{ca}(\theta_i, \dot{\theta}_i). \quad (35)$$

In the end, the movement of the CDRM is achieved by motor drivers and dc motors.

### B. Force Sensing and Admittance Control Frame

The CDRM's control system comprises trajectory planning, motor control, mode switching, and a display of the robot's actual situation. There are 15 dc motors in the system, and it must coordinate their position/velocity control while meeting the accuracy requirements of each motor. In order to satisfy the system's real-time requirements, we design a multilayer controller.

In the design of the control system, it is necessary to consider the real-time centralized acquisition and transmission of the signals of each sensor. In addition, the next path planning signal obtained by the algorithm needs to be transmitted to the hardware in time to fully guarantee the real-time performance of the CDRM system. The high-level controller can display the value of each sensor in real time and send the target position control signal in real time. Considering the above-mentioned design, the system includes a human-computer interaction layer, an algorithmic control layer, and a hardware layer, as shown in Fig. 9.

The trajectory planning, open-/closed-loop/admittance control, force-sensing algorithm, and other algorithms are included in the algorithmic control layer. With a frequency of 1 kHz, PID controllers provide independent position and velocity control for the 15 motors. High-level configuration and relevant data input/output (e.g., joint commands) are provided over robot operating system (ROS) topics. In order to sense the external force at the end-effector, the "force sensing" algorithm in the

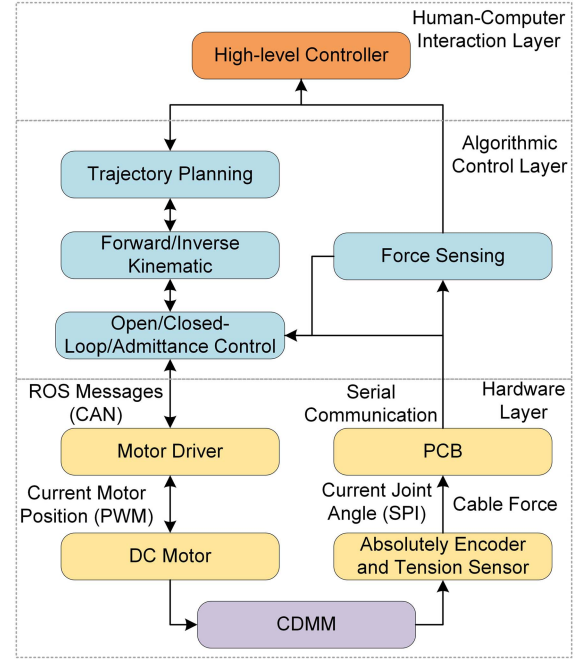


Fig. 9. High-level controller sends motor commands as ROS messages over controller area network (CAN) to the driven modules.

TABLE I  
MATERIAL AND GEOMETRIC PARAMETERS OF THE PROTOTYPE

| Symbol          | Property                          | Value                                         |
|-----------------|-----------------------------------|-----------------------------------------------|
| $m_L$           | Link Mass                         | 291.9 g                                       |
| $d$             | Distance between two surfaces     | 80 mm                                         |
| $r$             | Radius of disk                    | 50 mm                                         |
| $M$             | Mass matrix (for each joint)      | diag( $1 \cdot 10^{-4}$ , $1 \cdot 10^{-4}$ ) |
| $D$             | Damping matrix (for each joint)   | diag(0.01, 0.01)                              |
| $K$             | Stiffness matrix (for each joint) | diag(2, 2)                                    |
| $l$             | Link Length                       | 150 mm                                        |
| $S_{fm}$        | Stribeck friction                 | 1.5 N                                         |
| $K_v$           | Viscous coefficient               | 0.28                                          |
| $K_p, K_i, K_d$ | PID parameters                    | 200, 100, 2                                   |

algorithm layer utilizes sensor signals sent in real time by the hardware layer via the custom printed circuit board (PCB).

The hardware layer provides execution and feedback functions. The motor control signal of the algorithm layer is sent to the motor driver to drive the CDRM to execute the desired motion. In addition, there are a large number of motor encoders, absolute magnetic encoders, and tension sensors in the hardware layer to collect the current state. In order to improve the efficiency of signal acquisition, the PCB based on STM32 is designed to collect all the signals from the sensors in real time, store them centrally, and send them to the algorithm layer and human-computer interaction layer simultaneously.

## V. EXPERIMENT ANALYSIS

### A. Prototype and Experiment Setup

The overall construction and parameters of the prototype are shown in Fig. 10 and Table I. The modularly driven unit includes



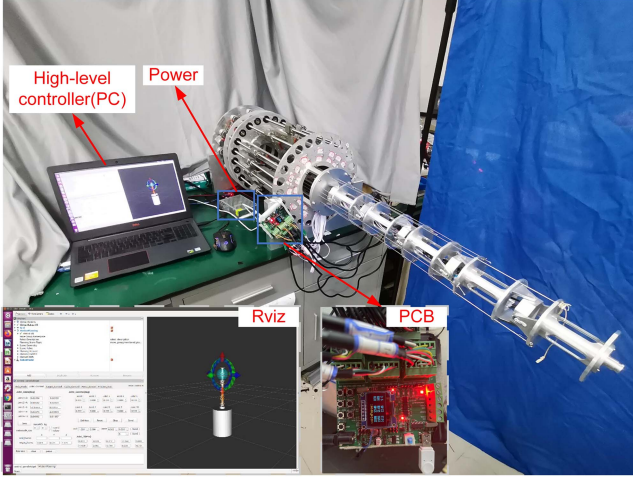


Fig. 10. Prototype and hardware system of CDRM.

a ball-screw type linear guide, a dc motor (RE25, Maxon motor), a gearbox (GP22, Maxon motor), and a servo drive. The joint encoder and the power amplifier signal line of the tension sensor are connected to the PCB, which is connected to the PC through the serial port.

According to the experiment requirements, we create ROS Kinect (Ubuntu 16.04) software for control, and the human-computer interaction interface is built with Rviz and QT4.2 as shown in Fig. 10.

### B. Estimation of Friction

Before validating the force-sensing method and admittance control, the main challenge in the experiments is to estimate the friction. As depicted in Fig. 11(a), a device for measuring the friction coefficient is designed and manufactured.

The first step of the experiment is to select the bending angle between the cable and the hole and drive the cable to move at a uniform speed. Then, the tensions  $F_{1,n,k}$  and  $F_{2,n,k}$  at both ends of the cable hole are measured, and the dynamic friction force  $f_{i,k}$  is calculated through the dynamic friction force model. The parameter values of the friction model are given in Table I, and the Coulomb equivalent coefficient  $\mu'$  polynomial of friction is given as follows:

$$\begin{aligned} \mu'(\theta_{b,n}) = & 3.48 \cdot 10^{-7} \theta_{b,n}^5 - 2.29 \cdot 10^{-5} \theta_{b,n}^4 + 4.64 \cdot 10^{-4} \theta_{b,n}^3 \\ & - 3.52 \cdot 10^{-3} \theta_{b,n}^2 - 3.33 \cdot 10^{-3} \theta_{b,n} + 0.004 \text{ (pull)} \end{aligned} \quad (36)$$

$$\begin{aligned} \mu'(\theta_{b,n}) = & -7.23 \cdot 10^{-8} \theta_{b,n}^5 + 3.37 \cdot 10^{-6} \theta_{b,n}^4 - 3.53 \cdot 10^{-5} \theta_{b,n}^3 \\ & + 1.77 \cdot 10^{-4} \theta_{b,n}^2 + 7.59 \cdot 10^{-3} \theta_{b,n} + 0.010 \text{ (loose)} \end{aligned} \quad (37)$$

Finally, the cable tension  $F_{1,n,k}$  and the real cable tension  $F_{2,n,k}$  are recorded and compared. The average values of the actual cable tension  $F_{2,n,k}$  in the four different motion states are 132.828, 163.23, 43.969, and 108.778 N, respectively, and the average errors of the calculated cable tension  $F_{2,n,k}$  are  $-4.309$ ,  $3.590$ ,  $-2.094$ , and  $-0.5587$  N. Obviously, a larger angle or

smaller tension results in a higher error and further affects the accuracy of force sensing. On the other hand, the 15 cables of the CDRM must have a certain initial value to ensure that the manipulator maintains the tension (30 N), which benefits the friction calculation.

### C. Force-Sensing Validation

We set up three experiments to validate the proposed force-sensing algorithm. In the first experiment, different weights are used at the end-effector of the CDRM in a static pose. Then, we add the dynamic disturbances at the six-axis  $F/T$  sensor (M8128B1, Sunrise Instruments) for force-sensing validation. Finally, the manipulator carries different weights and moves according to the planned trajectory in the third experiment, which also verify the proposed dynamics model.

Under the external payloads of 1 kg (9.8 N), 2 kg (19.6 N), and 3 kg (29.4 N), the average sensing forces are  $-9.68$ ,  $-18.971$ , and  $-28.511$  N, respectively, due to the negative direction of the load gravity along the  $z$ -axis in Fig. 12. Due to the noise of the tension sensor and joint encoder, there are also some sensing errors on the  $x$ -axis and  $y$ -axis, which are  $[-0.06, -0.207]$  N (1 kg),  $[-0.233, 0.413]$  N (2 kg), and  $[-0.576, 0.852]$  N (3 kg), respectively. The corresponding mean absolute errors of the three weights are 1.2%, 3.18%, and 2.96%, respectively, which are less than those reported in other research studies, 5%–20% [26]. This experiment shows that the force-sensing algorithm of the external force has a high accuracy, which can reach more than 96.8%, and the sensing is relatively smooth when the CDRM is at the static condition.

In the second experiment, we apply random force disturbance and use a six-axis  $F/T$  sensor to verify the dynamic accuracy of our force-sensing method. Before the experiment, we increase the cable tension to compensate for the influence of gravity from the six-axis  $F/T$  sensor, and the dataset is shown in Fig. 12(e). The root-mean-square deviations (RMSE) of the  $x$ -,  $y$ -, and  $z$ -axis are 0.892, 1.574, and 1.240 N, respectively. In addition, we found that even if high-frequency disturbances increase sensing error, admittance control can still be robust and stable with an appropriate damping parameter  $D$ , as illustrated in Fig. 12(f)-(g) and Table I.

In the third experiment, the robot moves according to the planned trajectory. The external force value is collected and calculated in real time, and the sensing results from the experiment are shown in Fig. 13.

The mean values of the sensing forces are  $-8.678$ ,  $-19.909$ , and  $-30.141$  N, respectively. When compared with the real external forces  $-9.8$ ,  $-19.6$ , and  $-29.4$  N, the mean absolute errors are 11.32%, 1.72%, and 2.60%. The comparison between estimation and measurement shows errors of 0.56% and 1.35% for  $x$ - and  $y$ -axis forces, respectively. The above-mentioned experiments show that the accuracy of dynamic force sensing reaches 94.78%, which is more accurate than the 0.2 N error at the tip level of 0.75 N in [25].

There are also still some methods to elevate the capabilities of CDRM. A notable noise in the results can be seen in Fig. 13(a) and (b), which are peaks in the  $x$ - and  $y$ -axis components relative

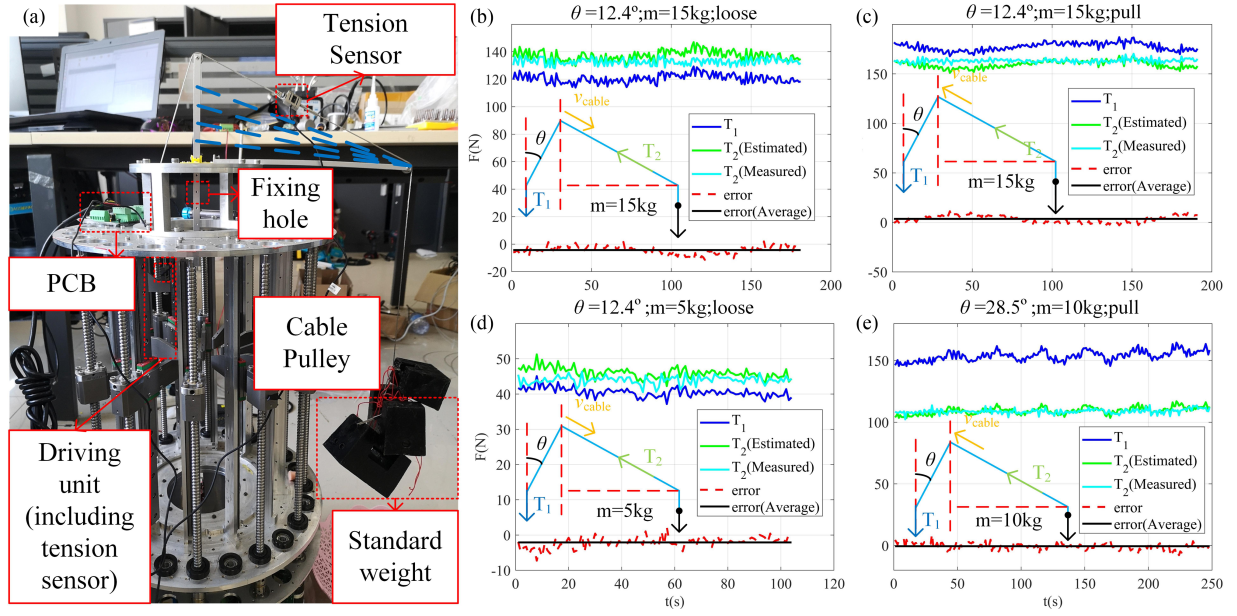


Fig. 11. (a) Friction coefficient measuring device. (b)–(e) Different configuration text results.

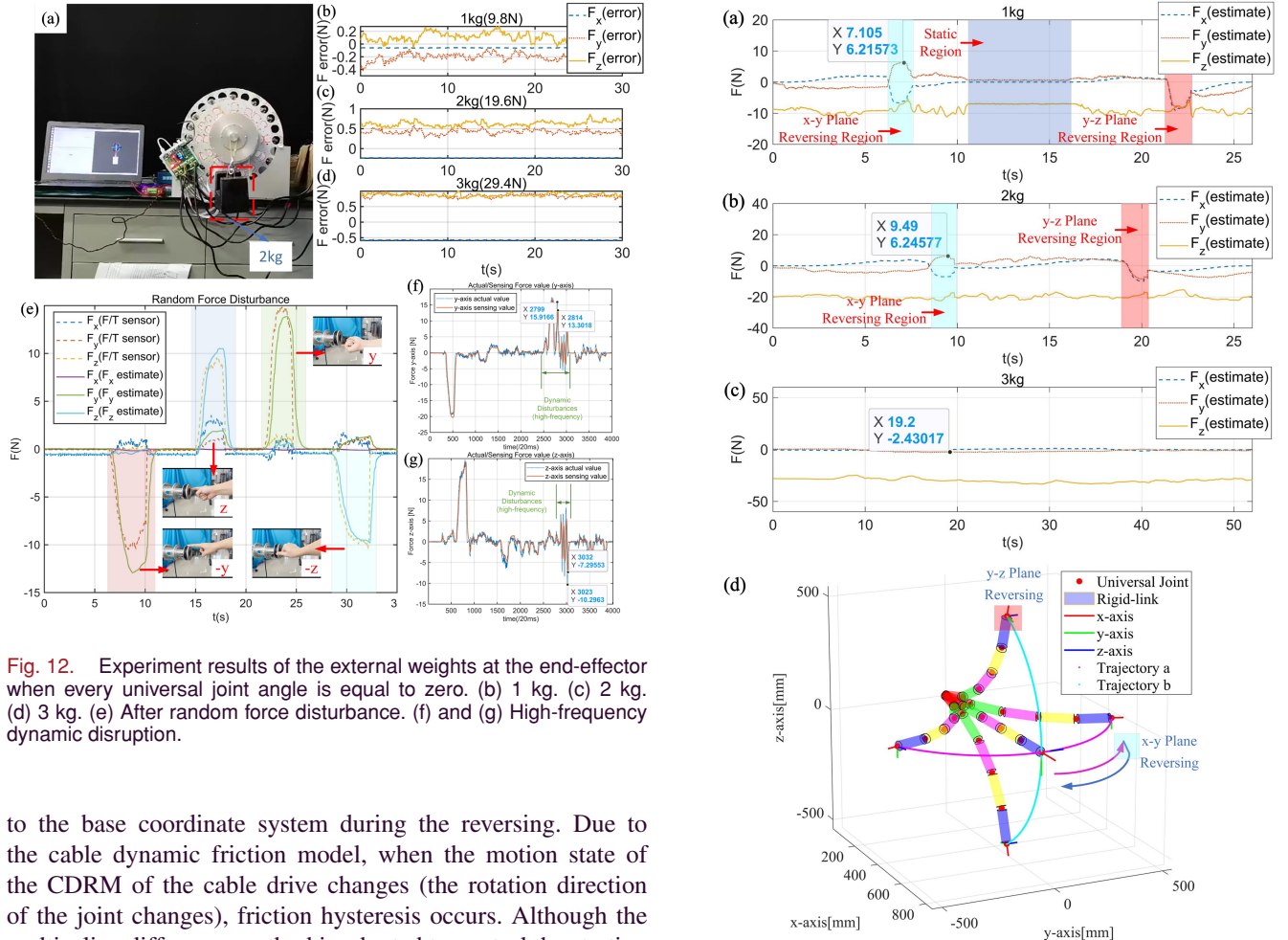
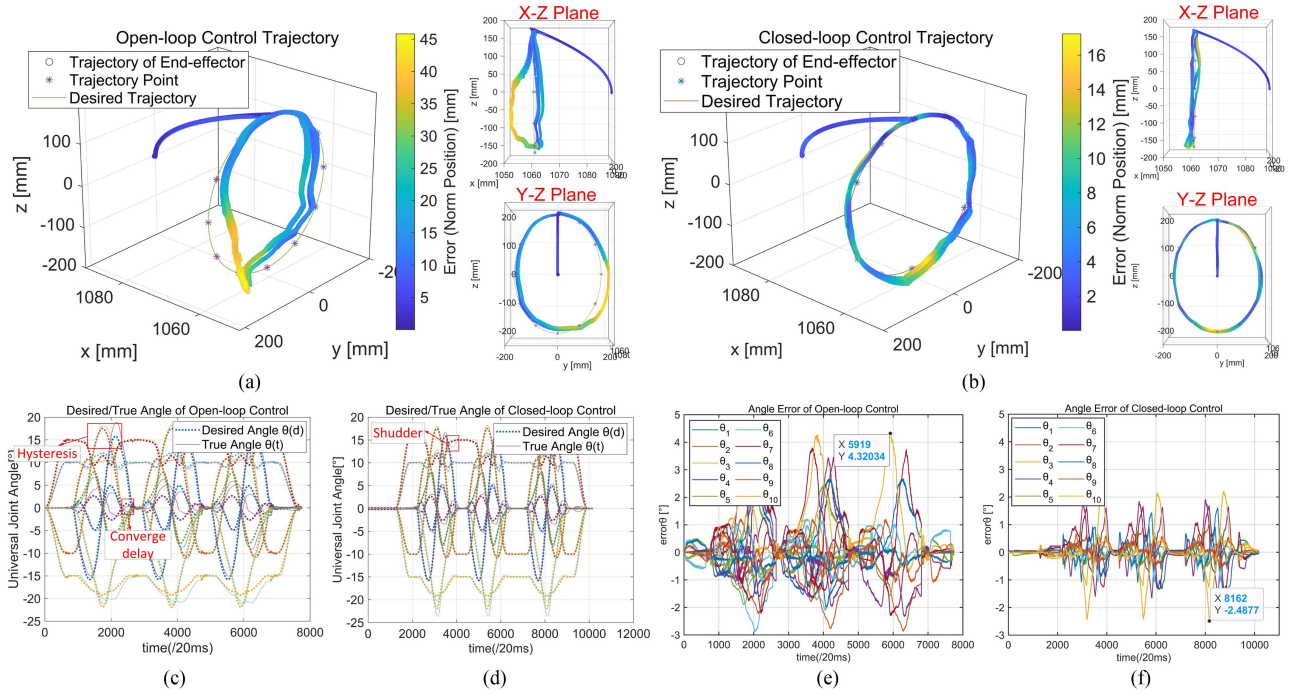


Fig. 12. Experiment results of the external weights at the end-effector when every universal joint angle is equal to zero. (b) 1 kg. (c) 2 kg. (d) 3 kg. (e) After random force disturbance. (f) and (g) High-frequency dynamic disruption.

to the base coordinate system during the reversing. Due to the cable dynamic friction model, when the motion state of the CDRM of the cable drive changes (the rotation direction of the joint changes), friction hysteresis occurs. Although the multispline difference method is adopted to control the starting and stopping speeds, the payload object has inertia due to its fast horizontal movement speed, which affects the accuracy of

Fig. 13. Experiment result of the force sensing along the certain path. (a) 1 kg. (b) 2 kg. (c) 3 kg.



**Fig. 14.** Tracking results for open-loop (left column) and closed-loop (right column) strategies for the CDRM. (a) and (b) Three-dimensional view and two different planes of the desired and true positions, and the color bar shows the value of position errors. (c) and (d) Desired/true angle of two different control strategies. (e) and (f) Ten joint angle errors of open-loop control and closed-loop control.

perception. Therefore, there is still some room for improvement in the treatment of friction hysteresis.

Considering the results given above, a notable result is that the accuracy of three-dimensional force sensing is distributed with respect to the velocity. Therefore, we increase the movement time of the third experiment (3 kg) to reduce the velocity [see Fig. 13(c)], and the noise of reversing motion is significantly reduced, which also validates our analysis.

#### D. Admittance Control Experiments

The universal joint closed-loop controller combines the advantages of feedforward and feedback control. The set of circle trajectories we use in the control experiments in this article have repeatability, and the joint motion ranges from  $-20^\circ$  to  $20^\circ$ . Tracking results are presented in Fig. 14, and the different colors indicate the position error between the current and the desired value. The maximum and average errors of tracking position are 45.81/17.67 mm and 17.23/4.31 mm, respectively. In addition to the track position, the joint angle following the open-loop control is delayed by more than 1 s; however, the closed-loop delay is less than 0.5 s under the same experimental conditions. Obviously, the closed-loop control can significantly eliminate the phenomena of hysteresis and convergence delay, further improving athletic performance and stability.

The maximum angle error of the open-loop control method is  $4.32^\circ$ , whereas the motion error of the position motion control method is still within  $2.49^\circ$ , and most of the angle error is within  $0.5^\circ$  in the closed-loop control, as shown in Fig. 14(e) and (f). In addition, in order to compare the performances of the two

control strategies under load conditions, 1- and 2-kg loads were applied, respectively, for the same motion. When the load is increased to 2 kg, the motion error of the manipulator reaches  $5.71^\circ$  in the open-loop motion mode, and the maximum motion error of the manipulator is  $2.64^\circ$  when the closed-loop control strategy is adopted. It can be seen that the closed-loop control can maintain smaller errors under external loads.

In order to verify the kinematics of the manipulator after using the high-precision closed-loop control algorithm, the positioning accuracy experiment is supplemented with OptiTrack. Starting from the origin, the CDRM moves more than 12 times on the planned trajectory and records the positions of markers in the world coordinate system. It can be seen that the RMSE is 6.72 mm, and the path-following error is 0.019 mm. Furthermore, we design the null-space motion of CDRM, and the average errors of position are 4.44 and 1.6 mm by the OptiTrack and encoder, respectively.

Obviously, even with closed-loop control, there are other ways to improve the performance of the manipulator. Higher precision absolute encoders will improve the angular accuracy of acquisition. On the other hand, some cables reversing movements will still make the joints follow the desired value badly, resulting in a partial shudder in the closed loop, as shown in Fig. 14(d). Lubrication is the only method to reduce effectively.

Another experiment is the admittance control experiment. The experiments are carried out to verify the effect that the end-effector of the CDRM moves with the force change in the admittance control mode. The  $M$ ,  $D$ , and  $K$  parameters of the admittance control mode are adjusted, and then the experimenter applies an external force to the end of the manipulator, as shown



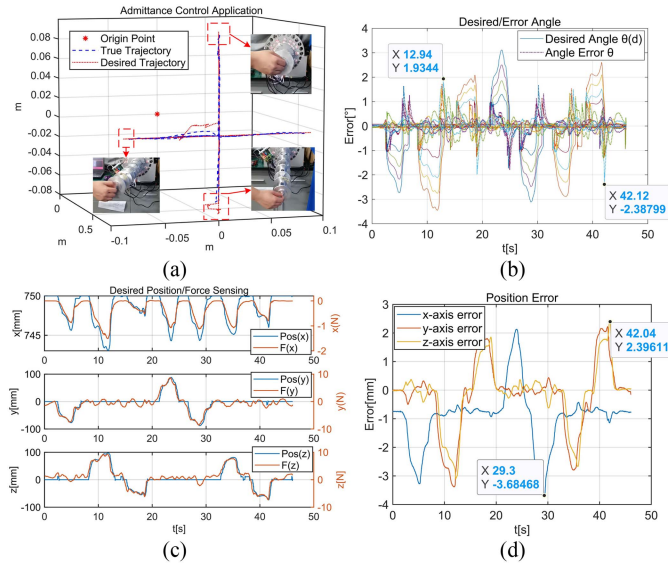


Fig. 15. Experiment results of the admittance control mode.

in Fig. 15(a). When the CDRM is moving, the calculated values of the positions, the universal joints, and the force sensing are measured in real time, as shown in Fig. 15(b)–(d).

According to the external force-sensing value, we can obtain the trajectory as shown in Fig. 15(a), and CDRM has a good following ability in the admittance control mode. Furthermore, the angle and position maximum errors occur at the same time, and this phenomenon is mainly caused by sudden changes in force caused by humans. The average error of the angles is  $0.212^\circ$ . Furthermore, according to the sensing data and encoder data, we find that the motion and external force have the same trend as shown in Fig. 15(c).

As depicted in Fig. 15(d), it can be seen that the errors of position are always less than 3.7 mm, and the RMSE is 2.14 mm. In contrast, the RMSE of the position error under the controller presented in [35] and [36] with dynamics uncertainties are 70.2 mm and 3.45 mm (total length at the level of 600 mm). In addition, the peak of the error is formed when the direction of the external force changes or when the direction of movement suddenly changes.

In summary, the force-sensing-based admittance control enables precise sensing of external forces at the end-effector and provides excellent external force tracking performance, making it an effective control strategy for driving the CDRM.

## VI. CONCLUSION

In this article, a novel force-sensing method based on the Newton–Euler method is proposed for the cable-driven redundant manipulator, and a CDRM experimental platform is further designed to validate the proposed method. By studying the high-precision motion control of the manipulator and the force sensing of the external force, the manipulator can achieve high-precision motion with closed-loop control and admittance control. The experiment results show that the dynamics and force-sensing models have more than 95% accuracy at high

speeds, and the perception accuracy is even higher at low speeds. On the other hand, the extended closed-loop control can effectively improve the accuracy of the CDRM and eliminate the hysteresis effect, which is based on the new structural design that includes ten absolute angle encoders. In general, the absolute position error is 4.31 mm, and the repeated error is less than 0.4 mm. In terms of application, we propose a novel admittance control based on force sensing without the six-axis force sensor and demonstrate excellent compliance performance. The applications of CDRM can be significantly expanded due to its variable stiffness and adaptation capabilities.

Our future work will focus on the force sensing of the entire manipulator without any external sensors on the links. In addition, we will design more intelligence control strategies for CDRM, which can further elevate the HRI capabilities.

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